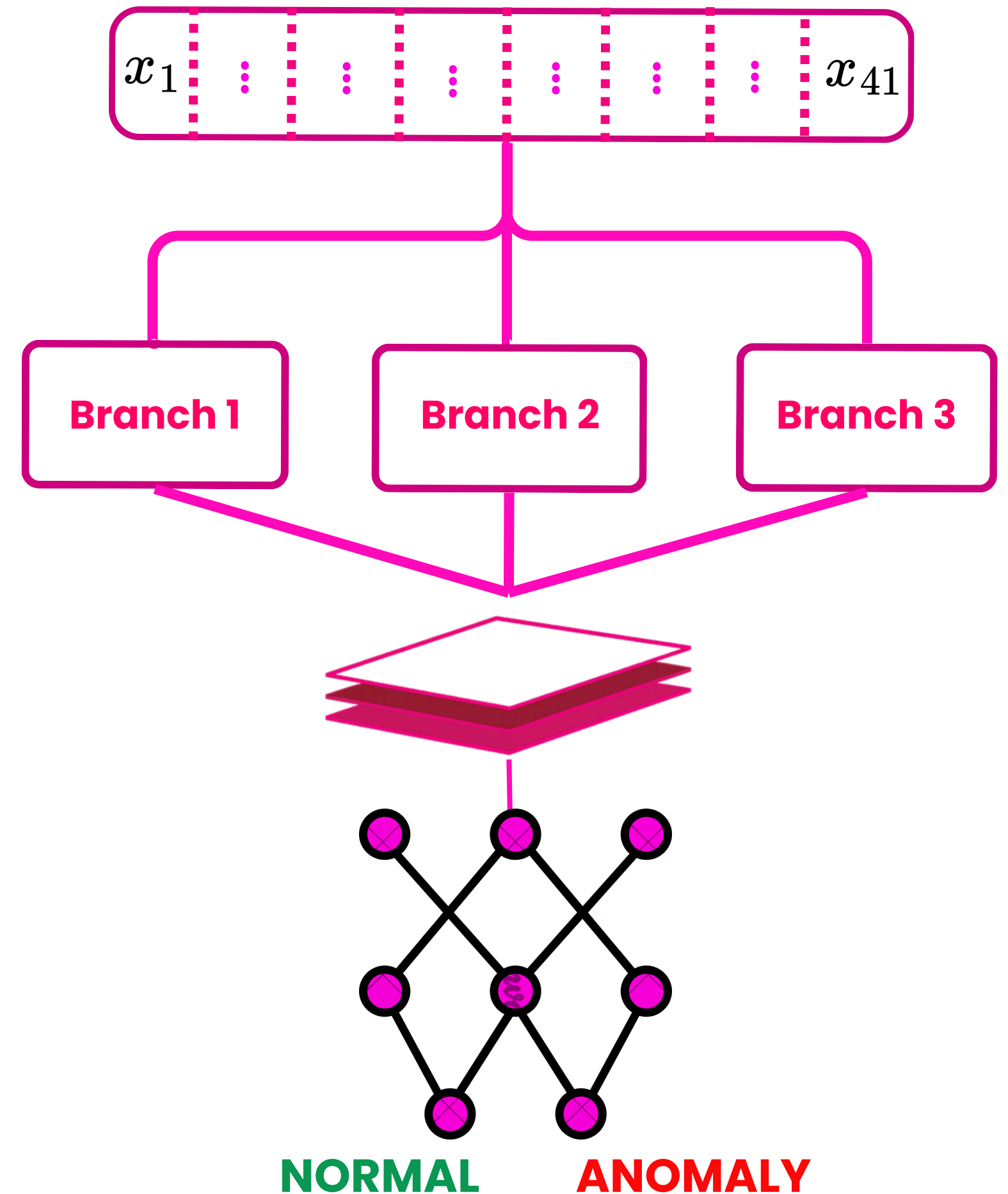


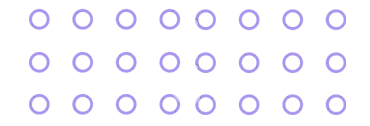
HYBRID AE-MSCNN WITH CALIBRATED FUSION FOR BINARY INTRUSION DETECTION ON NSL-KDD

Presented By :

Oussama BEN SASSI



PRESENTATION OUTLINE



- 01 THE ZERO-DAY PROBLEM.**
- 02 NSL-KDD DATASET OVERVIEW**
- 03 PROPOSED APPROACH PIPELINE, STANDARDIZATION, ENCODING AND BORDERLINE-SMOTE.**
- 04 AUTOENCODER AND MULTI-SCALE CNN CLASSIFIER.**
- 05 PERFORMANCE IN MERGED-SPLIT AND STRICT KDDTRAIN⁺ / KDDTEST⁺.**
- 06 COMPARATIVE TABLE WITH THE RELATED WORKS.**
- 07 DISCUSSION, LIMITATIONS AND CONCLUSION.**



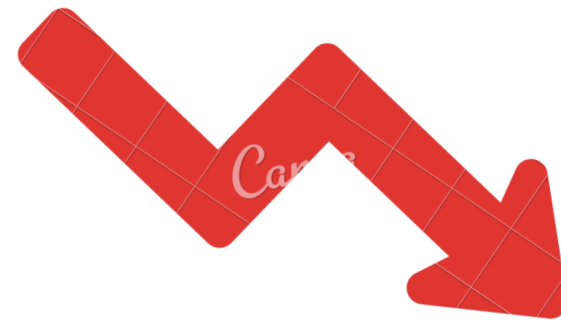
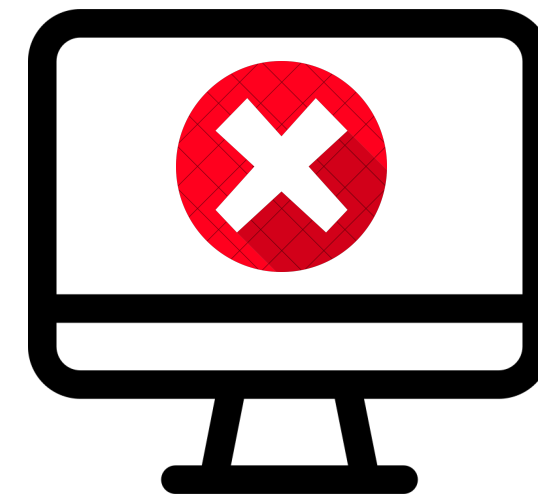
THE PROBLEM OF UNKNOWN ATTACKS

ZERO-DAY ATTACKS QUICKLY OUTPACE IDS TRAINED ON KNOWN THREATS.

CURRENT KNOWN ATTACKS

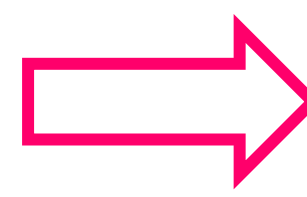


EMERGING UNKNOWN ATTACKS



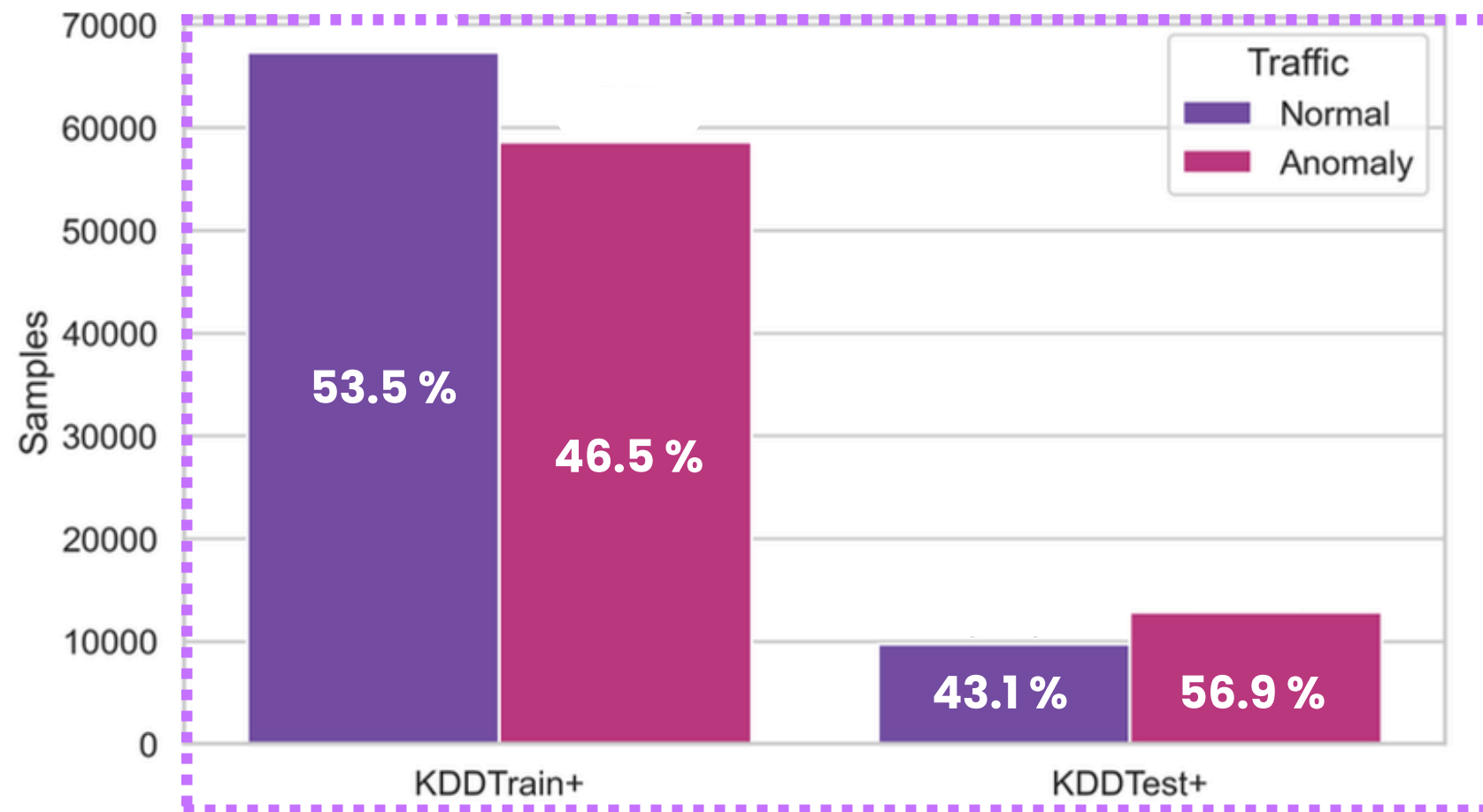
INITIAL DEPLOYMENT

YEARS LATER

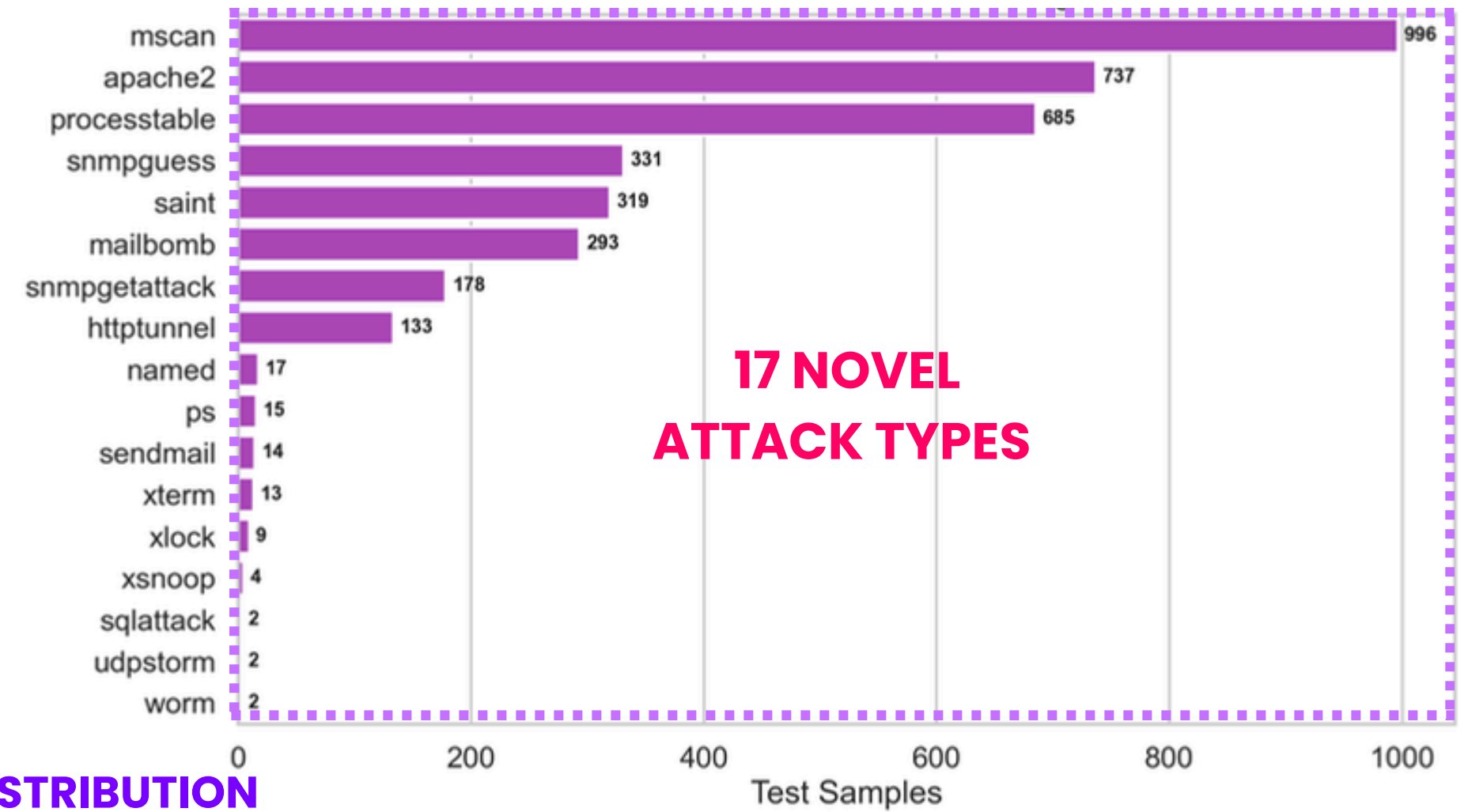
 **BUILD AN INTRUSION DETECTION SYSTEM THAT GENERALISES BEYOND ITS TRAINING DATA TO DETECT UNSEEN ATTACKS.**

NSL-KDD DATASET OVERVIEW

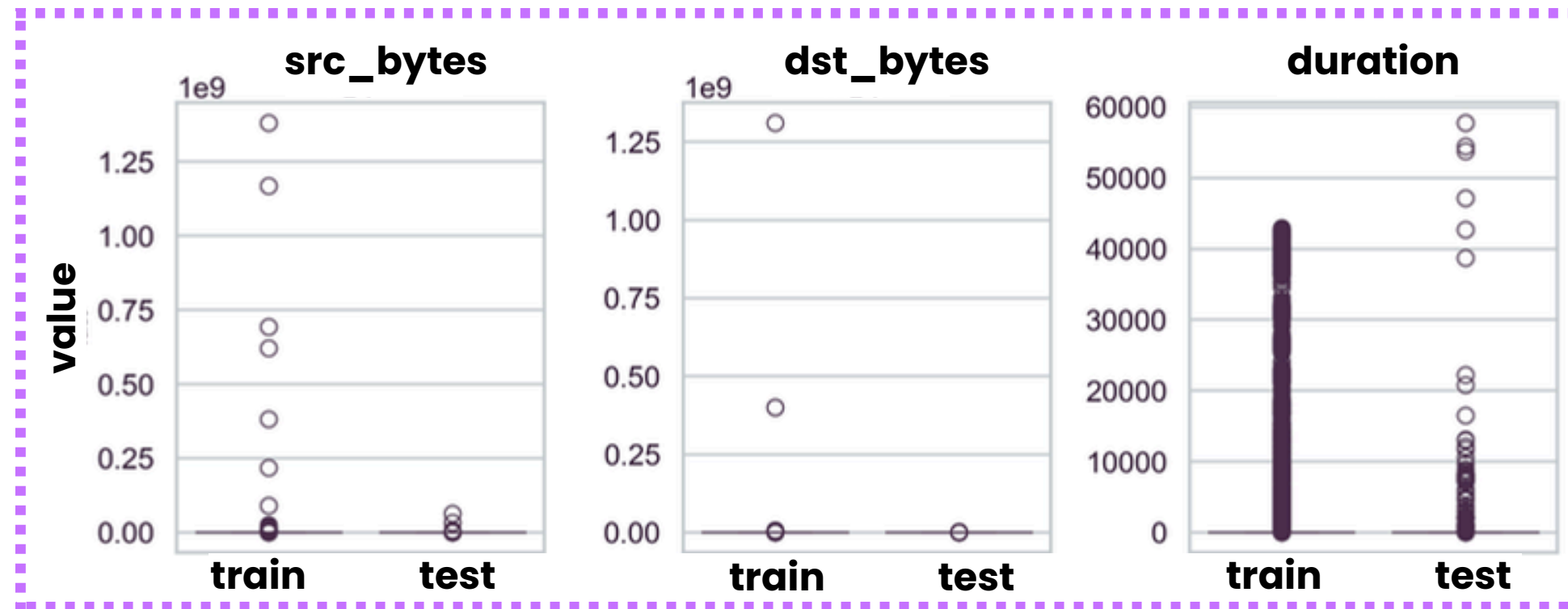
NORMAL VS ANOMALY TRAFFIC



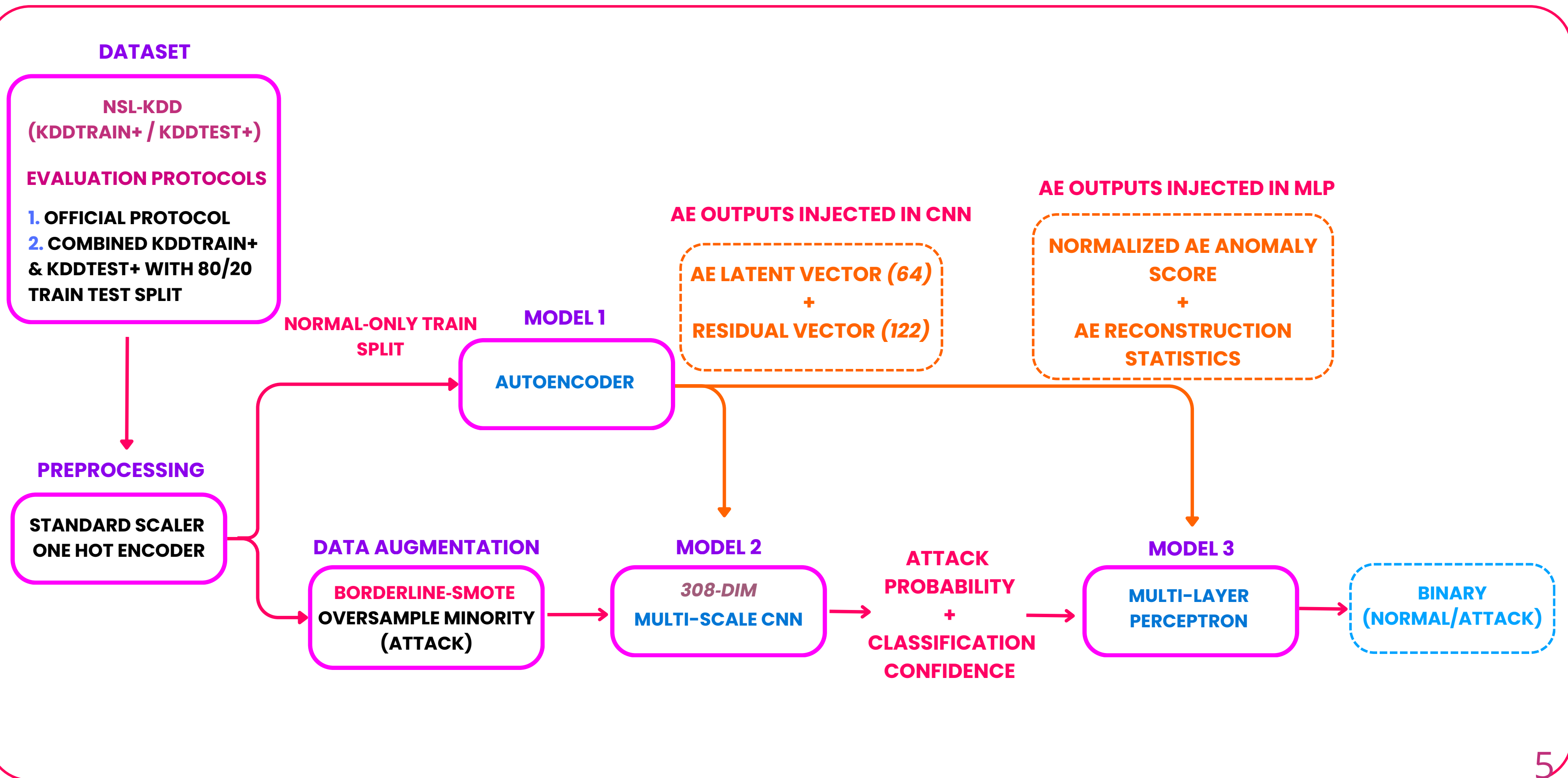
UNKNOWN ATTACK LABELS IN KDDTEST+



TRAIN VS TEST DISTRIBUTION

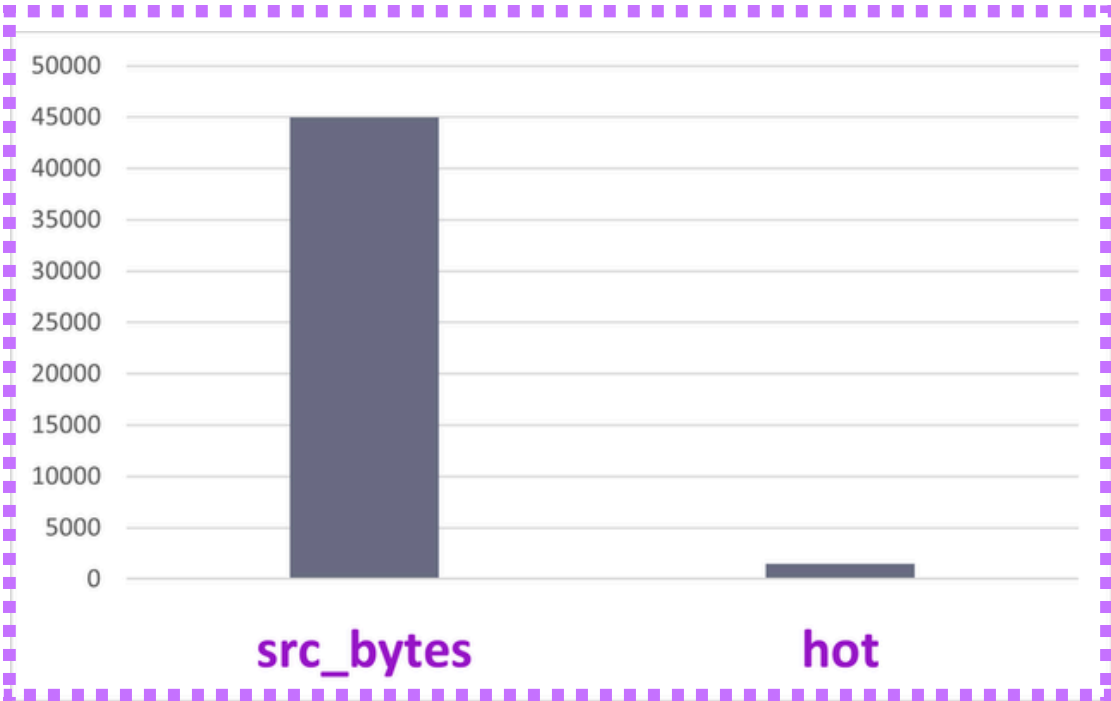


OUR APPROACH



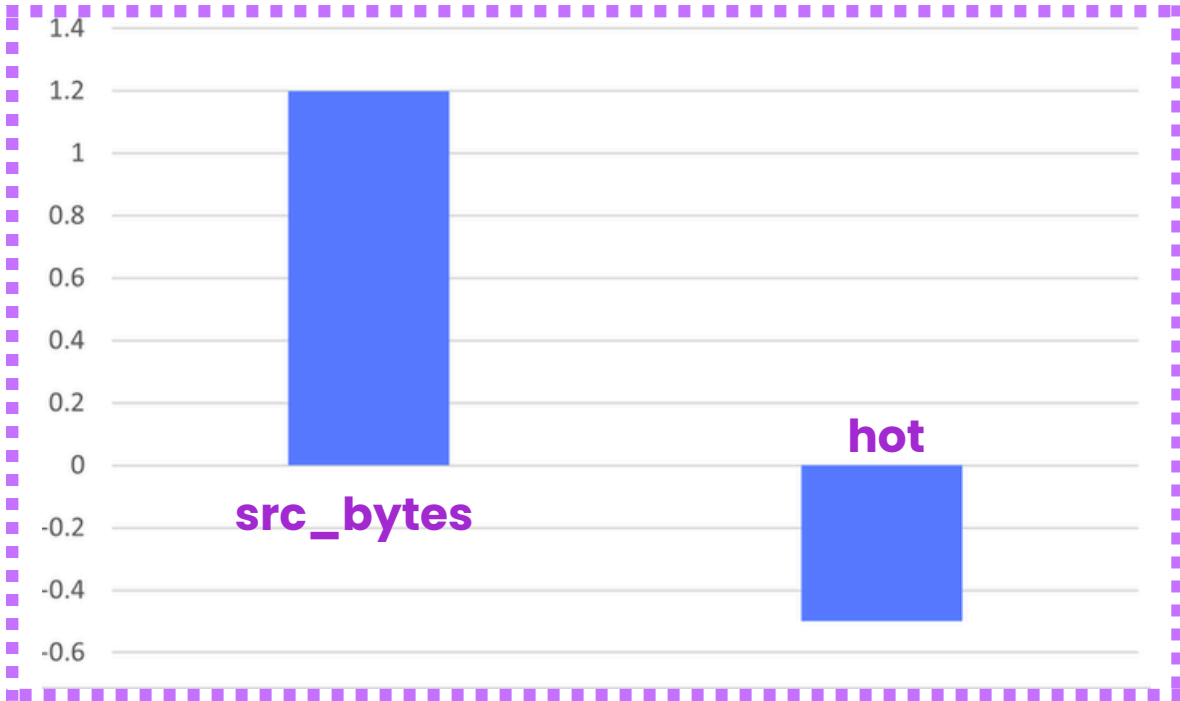
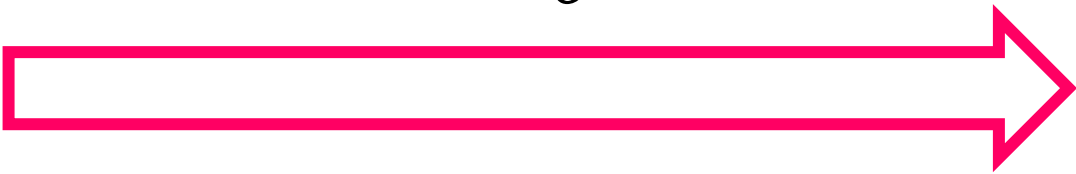
STANDARDIZATION & ENCODING

STANDARD SCALER ON NUMERICAL FEATURES



Z-SCORE NORMALIZATION [0, 1]

$$z = \frac{x - \mu}{\sigma}$$



ONE-HOT ENCODER ON CATEGORICAL FEATURES

Protocol_type
tcp
udp
icmp
tcp
....

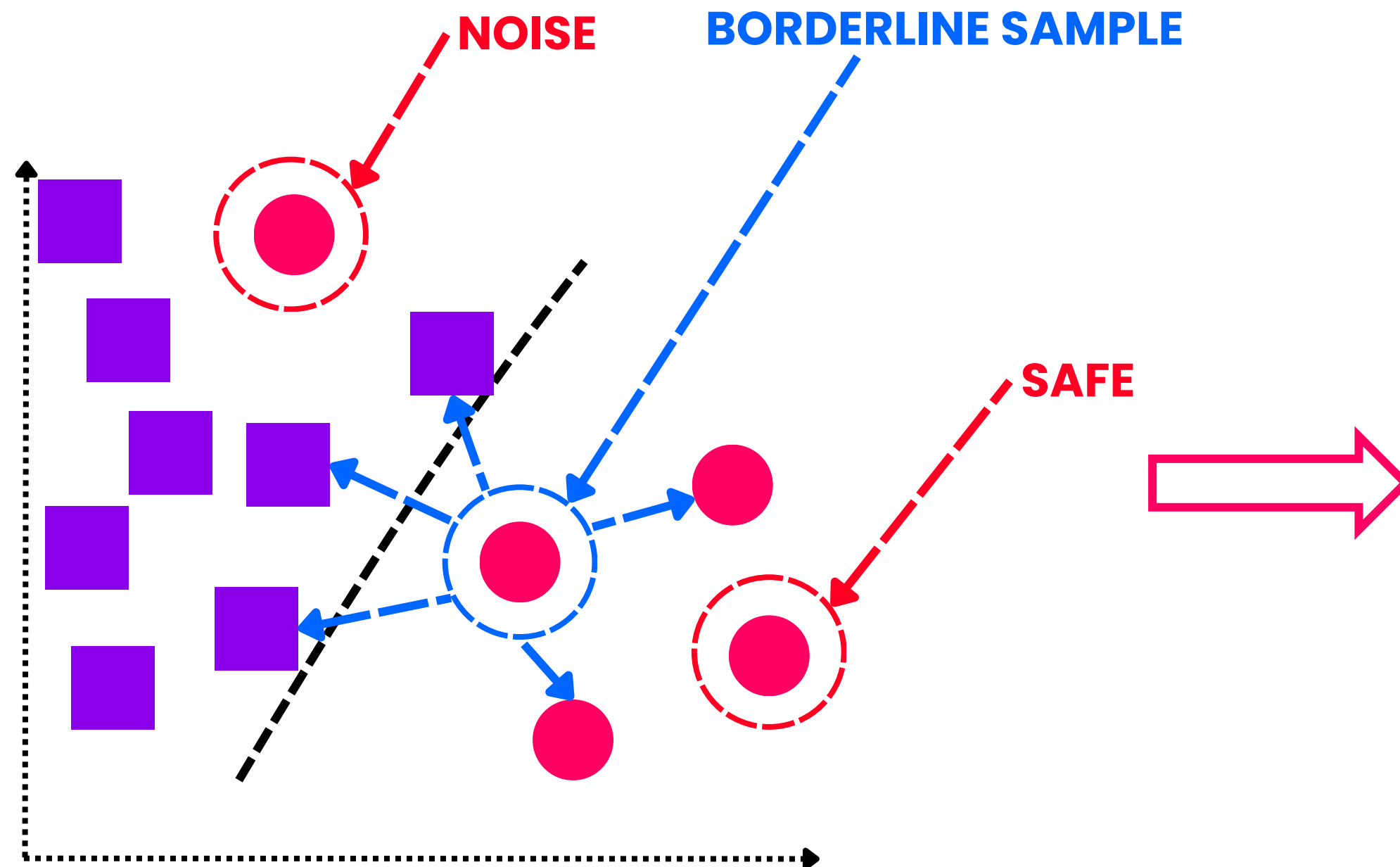
NON-ORDINAL MAPPING TO PREVENT FALSE NUMERICAL SCALING



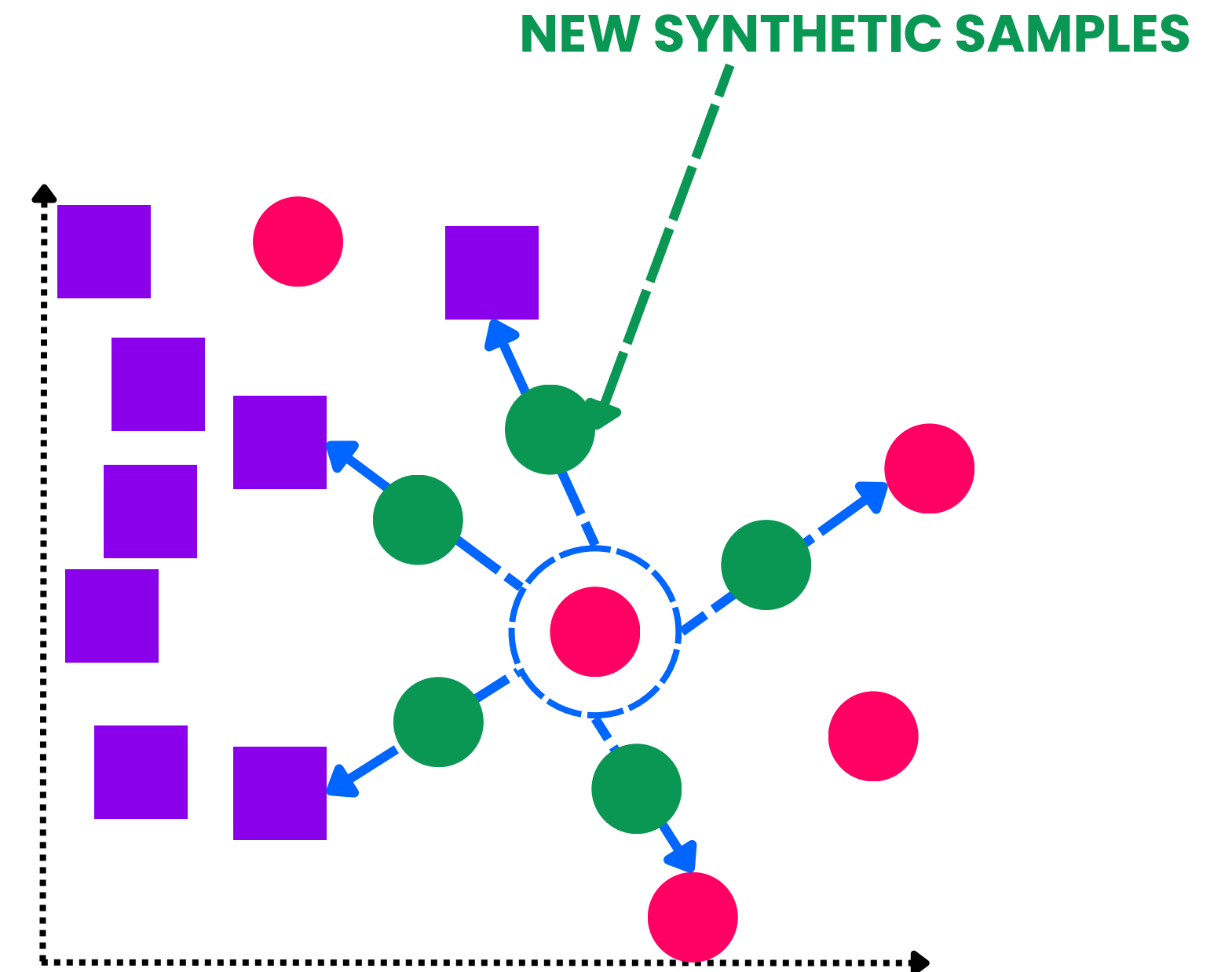
tcp	udp	icmp	tcp
1	0	0	0
0	1	0	0
0	0	1	0
1	0	0	0
....	...	...	...

AUGMENTATION WITH BORDERLINE-SMOTE

1. CATEGORIZE NEIGHBORS ($K = 5$)

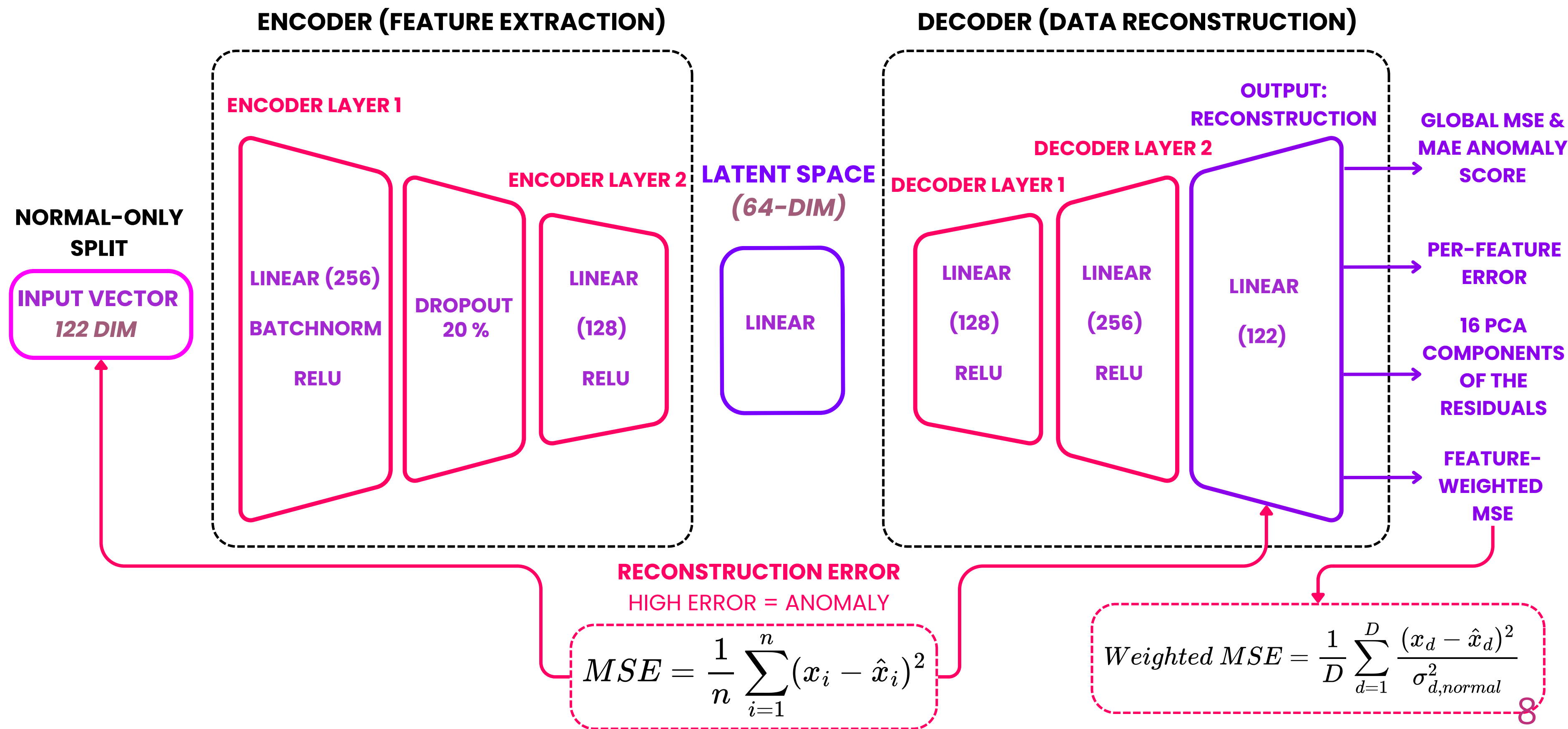


2. BALANCING THE MINORITY

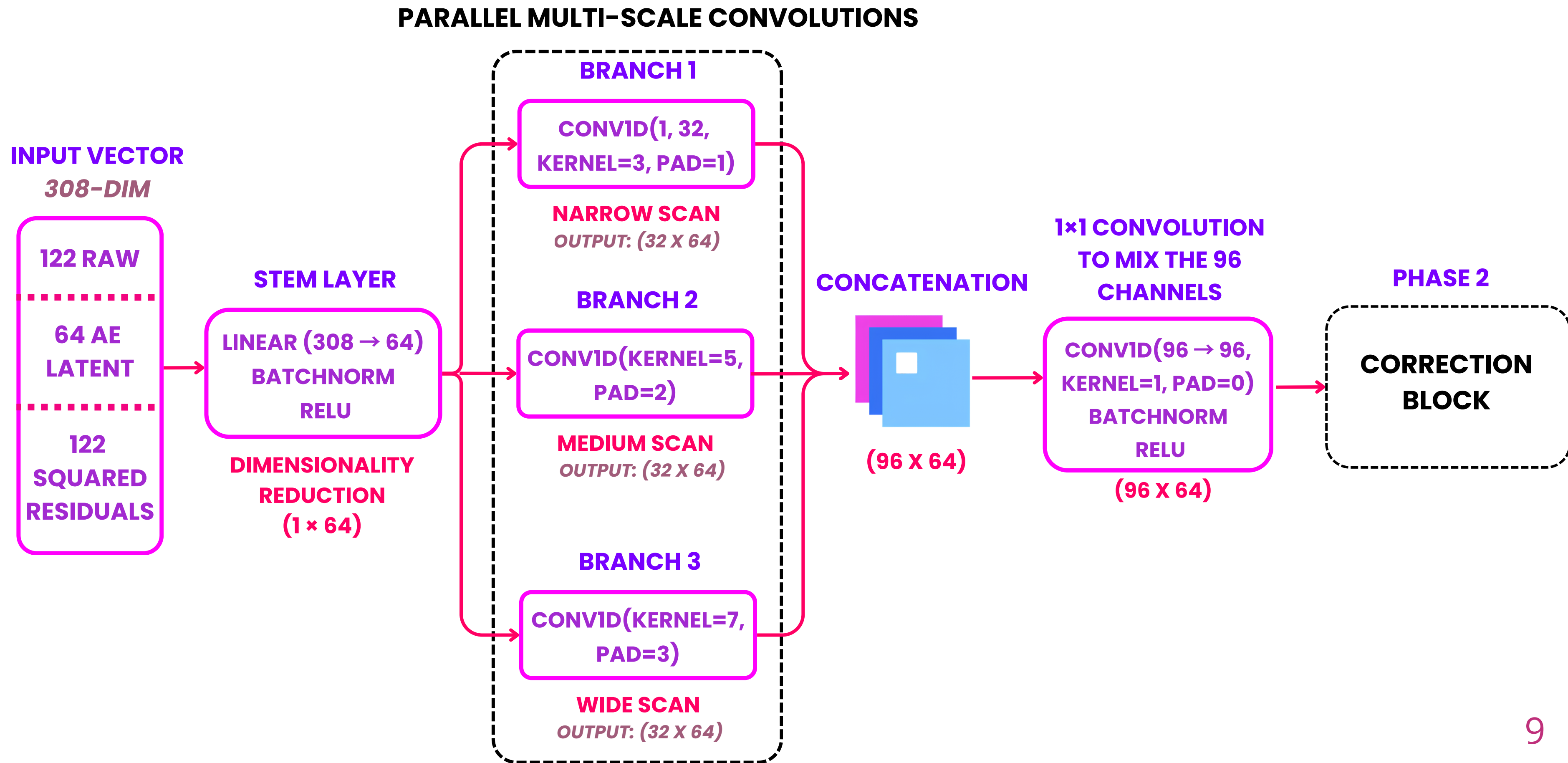


MINORITY CLASS NOW MATCHES
MAJORITY SIZE

UNSUPERVISED NORMALITY LEARNING USING AUTOENCODER

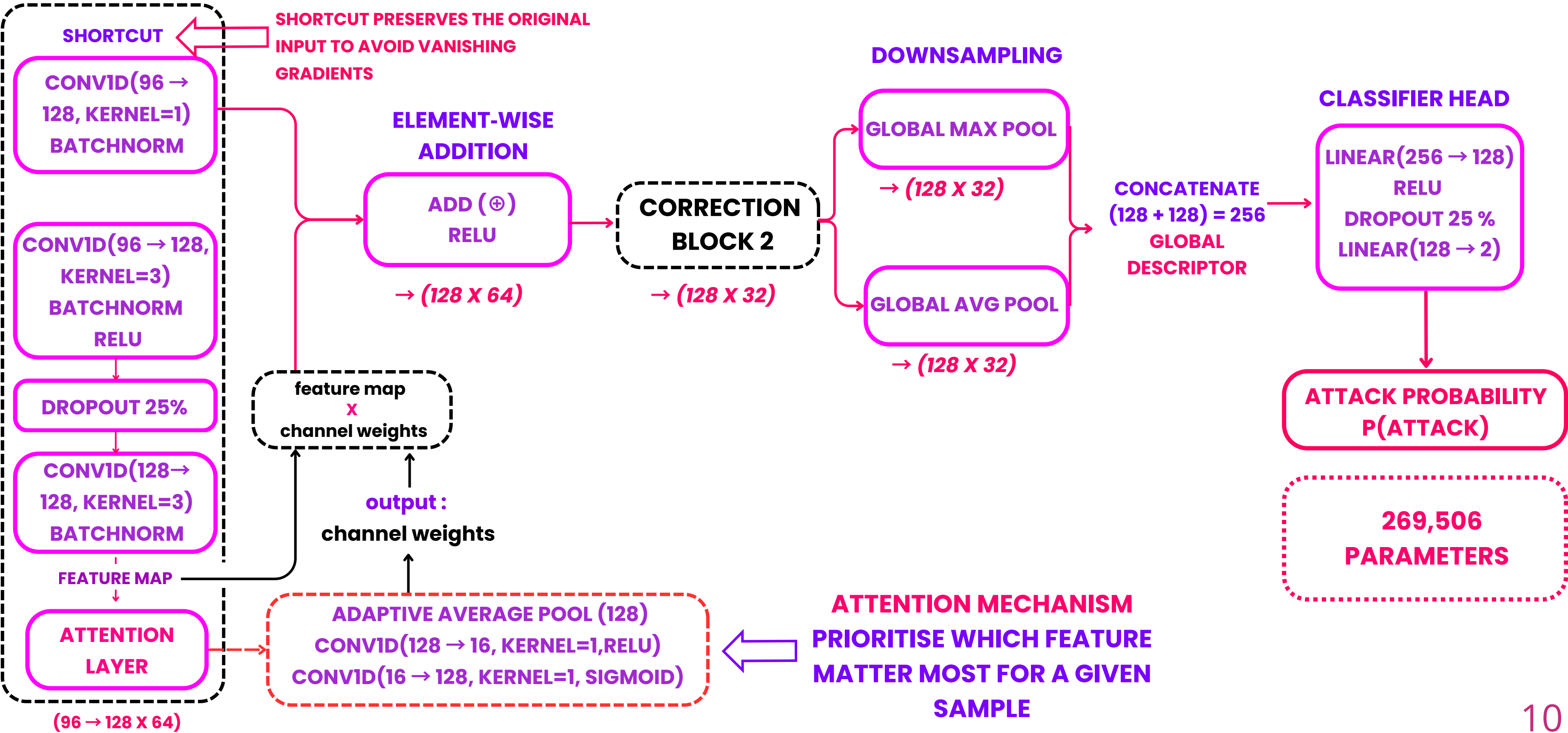


MULTI-SCALE 1D CNN CLASSIFIER : STEM & MULTI-SCALE BRANCHES



MULTI-SCALE 1D CNN CLASSIFIER : RESIDUAL & CLASSIFICATION HEAD

CORRECTION BLOCK 1



FUSION & DECISION WITH THE MLP

INPUT VECTOR
23-DIM

CNN ATTACK
PROBABILITY

CNN PREDICTION
ENTROPY

AE ANOMALY
SCORE

MAE & MSE
RECONSTRUCTION
ERROR

PER-FEATURE R.E

FEATURE-WEIGHTED
R.E

PCA-COMPRESSED
RESIDUAL

MLP (FUSION LAYER)

LINEAR (23→16)
BATCHNORM(16)
RELU
DROPOUT 30 %

LINEAR (16→1)
SIGMOID
→ ATTACK
PROBABILITY

LEARNS WHEN TO TRUST AE
VS. CNN AND PROVIDES THE
FINAL PROBABILITY.

ATTACK THRESHOLD SELECTION

For each threshold from [0,1]
(401 values) validation metrics
are calculated: (accuracy,
macro-F1, etc..) until condition is
met

CHOOSE THRESHOLD THAT
MAXIMISES ATTACK RECALL AND
NORMAL FPR ≤ 0.12

FINAL DECISION RULE :

IF ATTACK PROB $\geq$ THRESHOLD
ATTACK
ELSE :
NORMAL

MERGED-SPLIT PERFORMANCE (ZERO UNKNOWN ATTACKS)

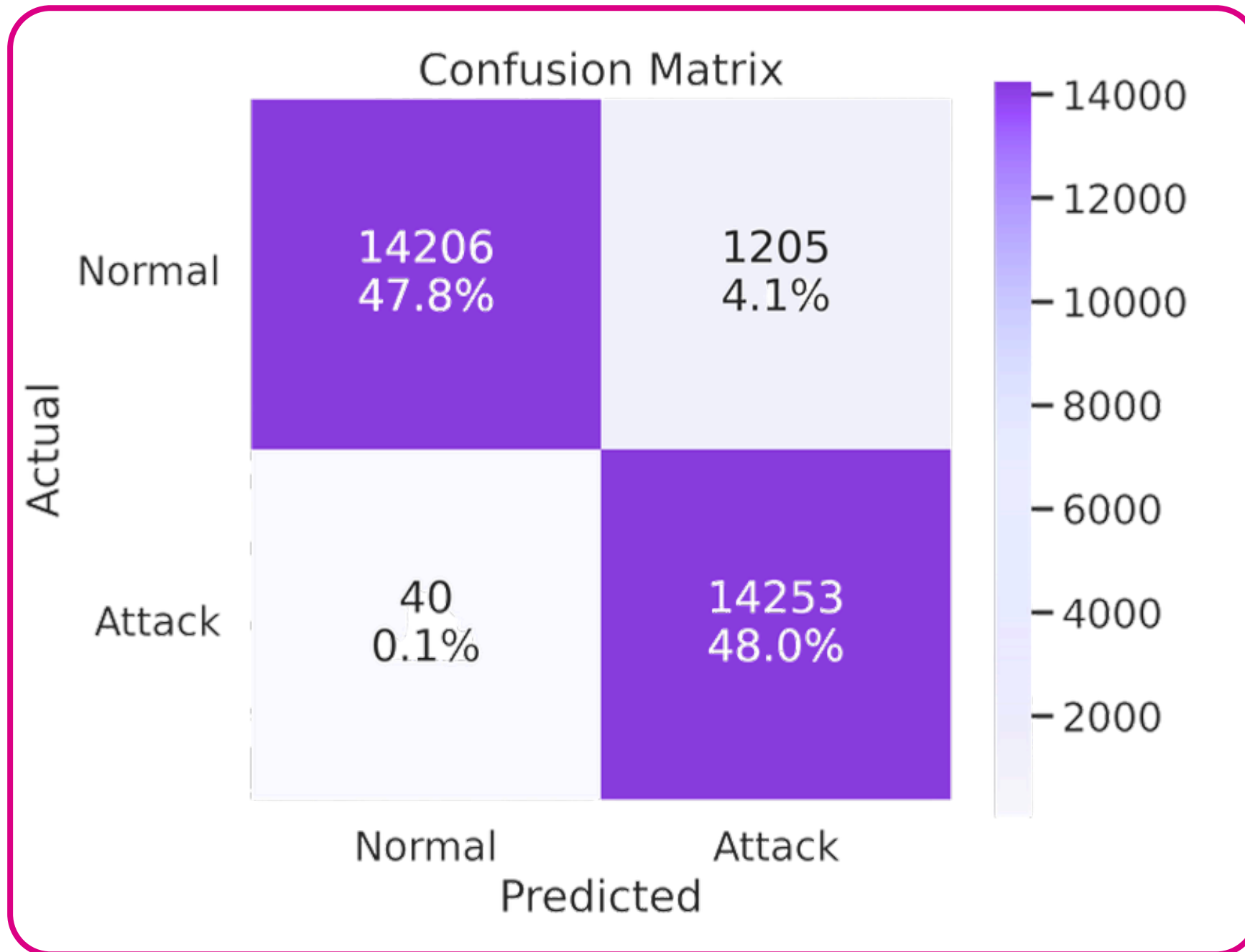
Metric	Standalone CNN	Hybrid (CNN + AE + MLP)
Accuracy	99.04%	95.81%
Macro F1	99.04%	95.81%
Attack Precision	98.69%	92.20%
Attack Recall	99.32%	99.72%
Attack F1	99.01%	95.82%
Normal FPR	1.22%	7.82%
ROC-AUC	99.93%	99.82%

STRICT KDDTRAIN+ / KDDTEST+ SPLIT (ZERO-DAY DETECTION)

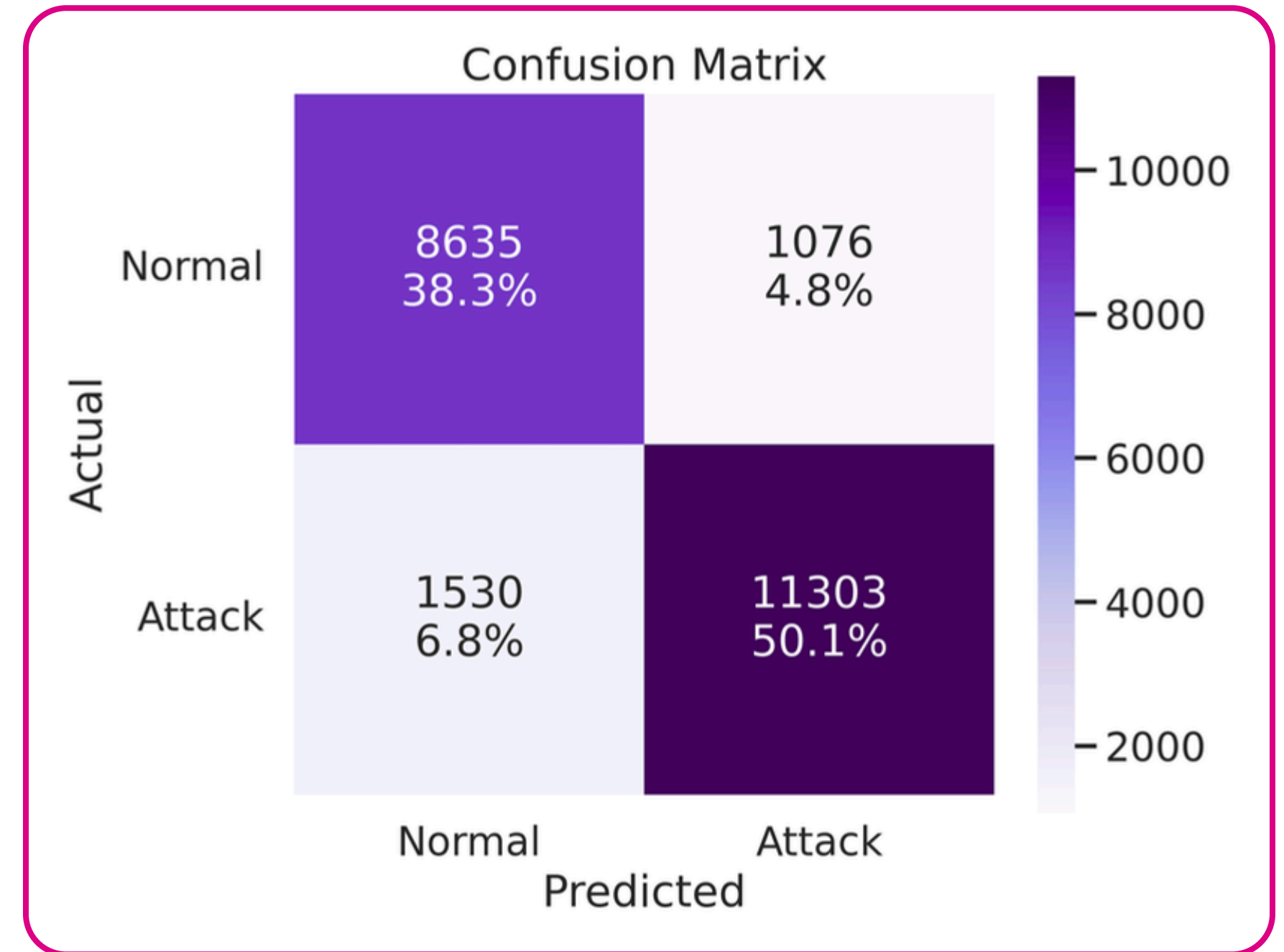
Metric	Standalone CNN	Hybrid (CNN + AE + MLP)
Accuracy	77.30%	88.44%
Known Subset Accuracy	87.02%	89.43%
Unknown Subset Accuracy	28.56%	83.47%
Known Attack Recall	76.43%	89.98%
Attack F1	75.79%	89.66%
Macro F1	77.21%	88.22%
Normal FPR	3.07%	11.08%
ROC-AUC	85.85%	95.72%

CONFUSION MATRICES

MERGED KDDTRAIN+ & KDDTEST+



OFFICIAL KDDTRAIN+ / KDDTEST+ SPLIT



➡ THE PERFORMANCE DROP IN ZERO-DAY SETTING SIGNALS THAT OUR SYSTEM STRUGGLES TO RECOGNISE ATTACKS IT HAS NEVER SEEN (1530 FALSE NEGATIVES)

COMPARATIVE TABLE - MERGED SPLITS

Study	Year	Method	Split	Accuracy
Elsayed et al.	2024	Optimised CNN	80/20	99.48 %
Anuja & Annrose	2025	AE-CNN-LSTM	70/30	99.8 %
Mo et al	2025	Sparse AE + CNN	5-fold cross-validation	98.36 %
Hindy et al.	2020	Deep Autoencoder	75/25 normal-only	92.96 %
Our Standalone MS-CNN	2026	Multi-scale 1D CNN	80/20 merge	99.04 %

COMPARATIVE TABLE - STRICT KDDTRAIN⁺ / KDDTEST⁺

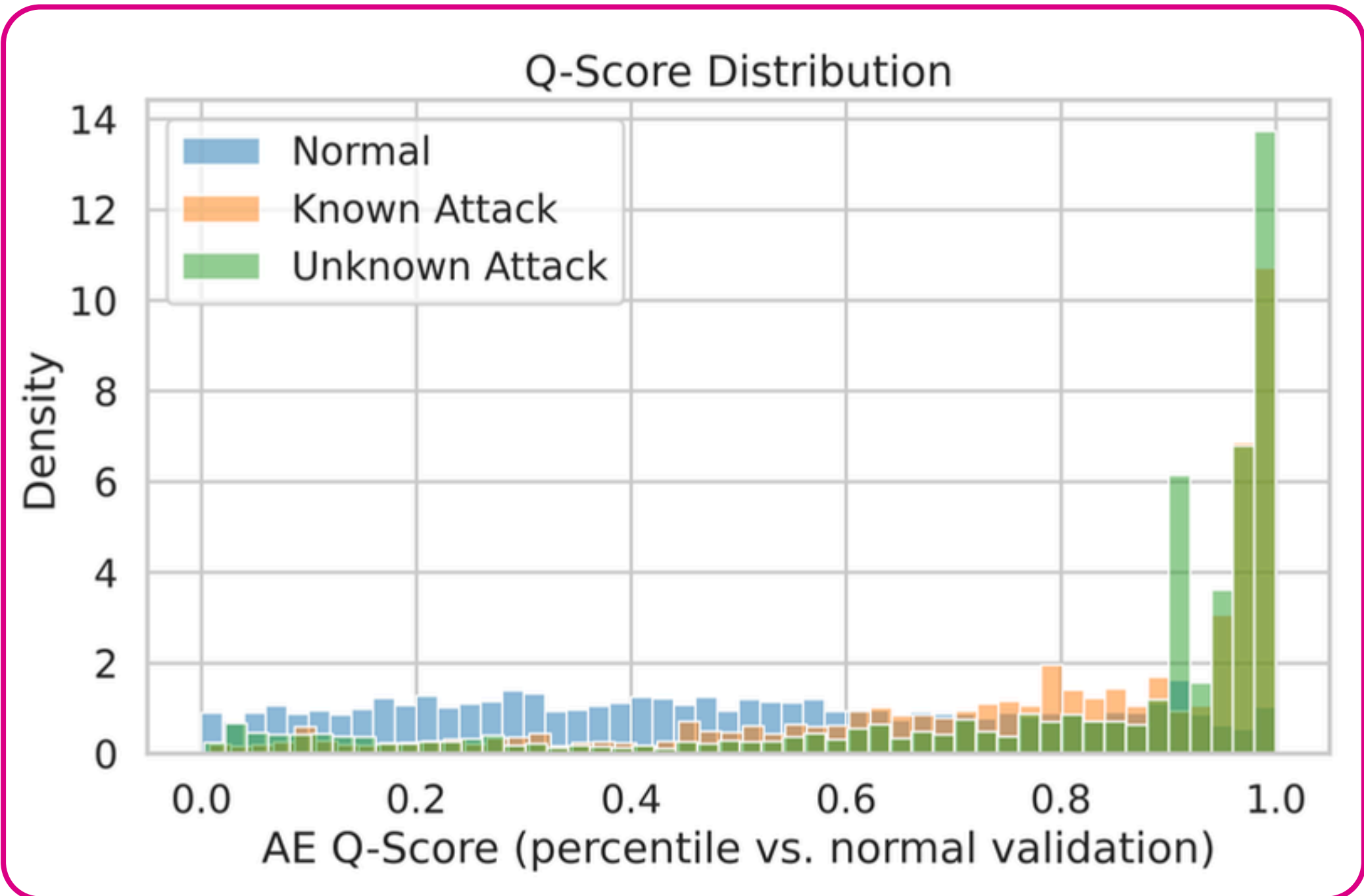
Study	Year	Method	Accuracy
Singh & Jang-Jaccard	2022	MSCNN-LSTM-AE	93.3 %
Fu et al.	2022	SAE-CNN-BiLSTM	90.7 %
Zeng	2025	CNN+Channel Attention Mechanism	91.1 %
Hussein & Répás	2026	AE-GMM-XGBoost	92.1 %
Our Hybrid Approach	2026	Hybrid AE-MSCNN + MLP	88.44 %

DISCUSSION & LIMITATIONS

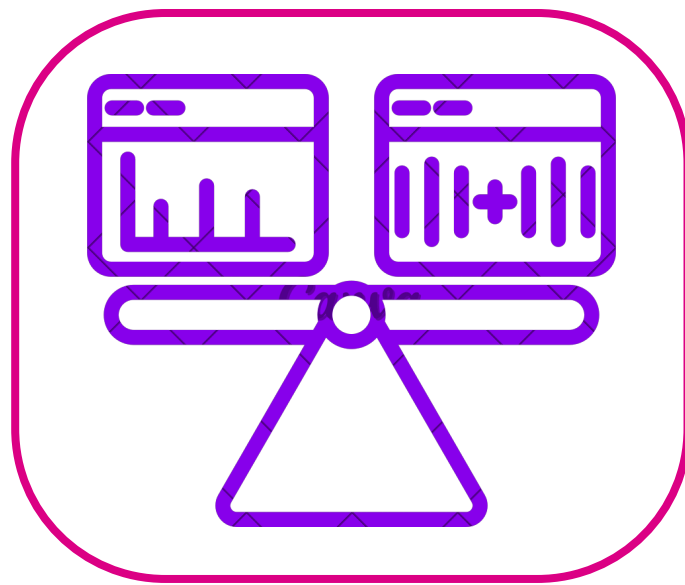
**THE AUTOENCODER FLAGS
RARE-BUT-NORMAL TRAFFIC AS
ANOMALOUS.**

**THE MLP FUSION GATE LIMITS THE
SYSTEM'S ABILITY FOR ZERO-DAY
DETECTION.**

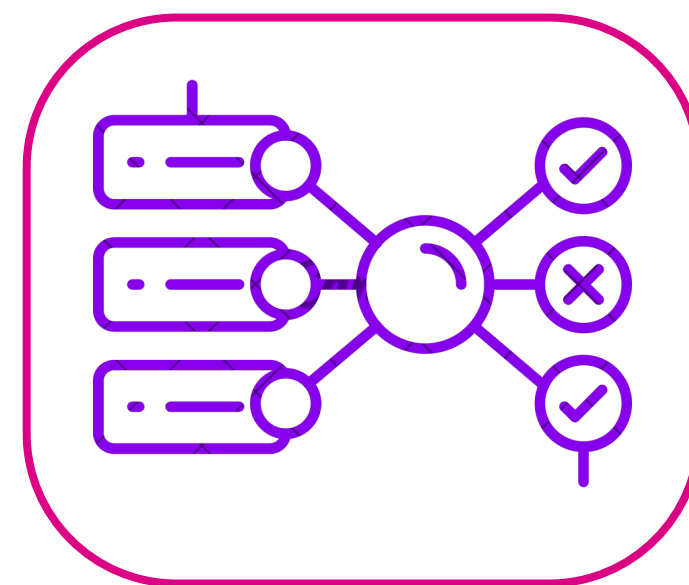
**OUR CURRENT APPROACH HAS NO
MECHANISM TO LEVERAGE TEMPORAL
CORRELATION IN NSLKDD DATASET.**



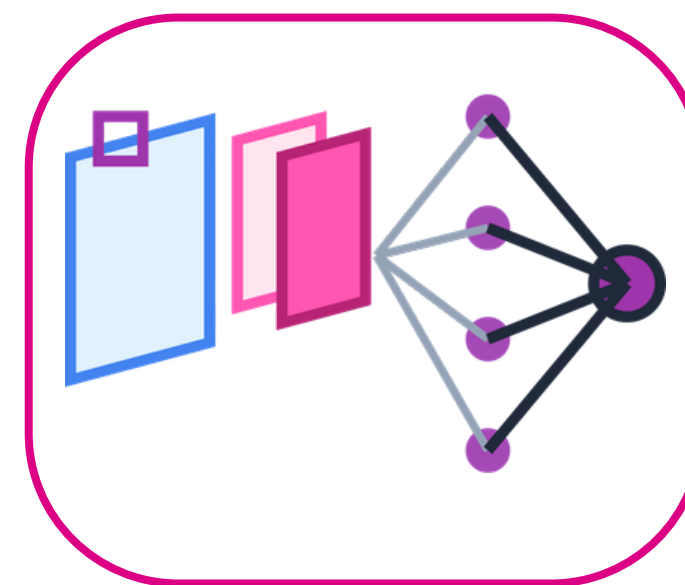
CONCLUSION



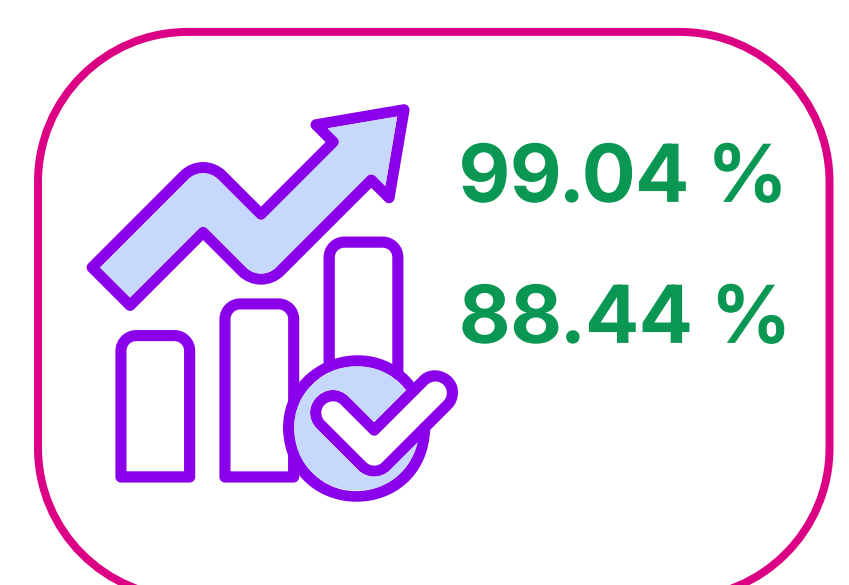
BORDERLINE SMOTE
DATA BALANCING



AUTOENCODER
FOR THE UNKNOWN



MULTI-SCALE CNN
POWERFUL CLASSIFIER



HIGH GENERALIZATION
COMPETITIVE PERFORMANCE

THANK YOU FOR YOUR TIME AND ATTENTION.

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